SOLAR▸SWARM

Master Index
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Solar Swarm — Master Documentation Index

Solar Swarm is a two-system platform built to generate recurring revenue from Australian solar SMEs using AI automation.

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Target: $1,500–2,000 AUD/month retainer per solar client

Margin: 80–90% (total costs ~$400 AUD/month)

What the System Does

System 1 — Autonomous Agent Swarm

A 3-tier hierarchy of AI agents that continuously generates, evaluates, and runs business experiments. The General (Tier 1) creates experiment ideas via GPT-4o, scores them using a 4-component confidence system, runs them through a red team critic, then routes them: auto-proceed, human gate, or auto-kill. Budget is allocated using a 25% Fractional Kelly Criterion. Agents communicate via a SQLite-backed message bus. Results feed back as pheromone signals that bias future experiment generation.

System 2 — Solar Sales Automation

A production CRM automation layer targeting Australian solar companies with 5–15 salespeople. When a lead arrives via GoHighLevel webhook, the system qualifies them with GPT-4o (score 1–10), triggers outbound AI voice calls for hot leads via Retell AI, updates the GHL pipeline, sends Slack alerts, and produces weekly performance reports.

ASCII Architecture Diagram

```
┌──────────────────────────────────────────────────────────────────────┐
│                          SOLAR SWARM                                 │
│                                                                      │
│  ┌──────────────────────────────────────────────────────────────┐    │
│  │                 SYSTEM 1 — AGENT SWARM                       │    │
│  │                                                              │    │
│  │  TIER 1        ┌──────────────────┐                          │    │
│  │                │   master_agent   │  (The General)           │    │
│  │                │   every 6h       │                          │    │
│  │                └──────────┬───────┘                          │    │
│  │                           │ generates + routes experiments   │    │
│  │  TIER 2        ┌──────────┴──────────────────────┐          │    │
│  │                │  research_agent │ content_agent  │          │    │
│  │                │  analytics_agent                 │          │    │
│  │                └──────────┬──────────────────────┘          │    │
│  │                           │ spawns workers                   │    │
│  │  TIER 3        ┌──────────┴──────────────────────┐          │    │
│  │                │ qualification │ proposal │ solar │          │    │
│  │                │ research │ report │ red_team      │          │    │
│  │                └──────────────────────────────────┘          │    │
│  │                                                              │    │
│  │  SUPPORT     scout_agent (daily 08:00 UTC)                   │    │
│  │              mutation_engine (Monday 22:30 UTC)              │    │
│  │              ab_tester (daily 10:00 UTC)                     │    │
│  └──────────────────────────────────────────────────────────────┘    │
│                                                                      │
│  ┌──────────────────────────────────────────────────────────────┐    │
│  │              SYSTEM 2 — SOLAR SALES AUTOMATION               │    │
│  │                                                              │    │
│  │  GHL Webhook → qualify lead → score 1–10 → route             │    │
│  │  score ≥ 7 → Retell outbound call + Slack alert              │    │
│  │  post-call → GPT extract → GHL update → task                 │    │
│  └──────────────────────────────────────────────────────────────┘    │
│                                                                      │
│  ┌──────────────────────────────────────────────────────────────┐    │
│  │                 SHARED INFRASTRUCTURE                        │    │
│  │                                                              │    │
│  │  Memory:  Hot (SQLite) → Warm (JSON) → Cold (ledger)         │    │
│  │  Capital: Kelly engine → Portfolio → Circuit breaker         │    │
│  │  Bus:     message_bus (SQLite, priority queues)              │    │
│  │  Data:    web_scraper, api_poller, social_signal             │    │
│  │  KG:      kg_entities / kg_relationships                     │    │
│  └──────────────────────────────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────┘
```

Quick Start

```
# 1. Clone and enter the project
cd Solar-Concept-main

# 2. Run the setup script (installs deps, creates .env, initialises DB)
bash setup.sh

# 3. Edit .env with your API keys (minimum: OPENAI_API_KEY)
nano .env

# 4. Start all services
python main.py

# 5. (Optional) Start the swarm board UI
cd swarm-board
npm install
npm run dev
# Opens at http://localhost:5173
```

Port Map

PORT	SERVICE	FILE
5000	Human Gate API (approve/reject experiments)	`api/human_gate.py`
5001	GHL Webhook Server (inbound lead events)	`webhooks/ghl_handler.py`
5002	Voice AI Webhook (Retell + ElevenLabs)	`voice/call_handler.py`
5003	Dashboard API (swarm-board live feed)	`api/dashboard_api.py`
5173	Swarm Board React App (Vite dev server)	`swarm-board/`

Environment Variables Reference

VARIABLE	REQUIRED	DEFAULT	DESCRIPTION
`OPENAI_API_KEY`	Yes	—	GPT-4o key. System runs in mock mode without it.
`GHL_API_KEY`	Yes (for CRM)	—	GoHighLevel private API key
`GHL_LOCATION_ID`	Yes (for CRM)	—	GHL sub-account location ID
`GHL_PIPELINE_ID`	No	—	GHL pipeline ID for stage sync
`GHL_WEBHOOK_SECRET`	No	—	Validates inbound GHL webhook signatures
`SLACK_WEBHOOK_URL`	No	—	Incoming webhook URL for Slack alerts
`SLACK_BOT_TOKEN`	No	—	xoxb- token for Slack Web API
`SLACK_SIGNING_SECRET`	No	—	Verifies interactive Slack button payloads
`SLACK_DEFAULT_CHANNEL`	No	`#swarm-alerts`	Default Slack channel
`WEEKLY_BUDGET_AUD`	No	`500`	Total weekly experiment budget in AUD
`GATE_API_KEY`	No	—	Bearer token to protect human gate endpoints
`RETELL_API_KEY`	No	—	Retell AI API key for voice calls
`RETELL_DEFAULT_VOICE_ID`	No	`11labs-Adrian`	Default Retell/ElevenLabs voice
`ELEVENLABS_API_KEY`	No	—	ElevenLabs API key
`ELEVENLABS_DEFAULT_VOICE`	No	—	ElevenLabs voice ID
`PORT_HUMAN_GATE`	No	`5000`	Port for human gate Flask app
`PORT_GHL_WEBHOOKS`	No	`5001`	Port for GHL webhook Flask app

VARIABLE	REQUIRED	DEFAULT	DESCRIPTION
`PORT_VOICE_WEBHOOK`	No	`5002`	Port for voice AI Flask app
`PORT_DASHBOARD_API`	No	`5003`	Port for dashboard API Flask app
`VOICE_WEBHOOK_BASE_URL`	No	`http://localhost:5002`	Public URL Retell can reach
`TRANSFER_PHONE`	No	—	Phone number for human transfer calls
`DEFAULT_CLIENT_ID`	No	`default`	Fallback client ID
`DATABASE_PATH`	No	`swarm.db`	SQLite database file path
`LOG_LEVEL`	No	`INFO`	Python logging level
`HUBSPOT_API_KEY`	No	—	HubSpot private app token
`SALESFORCE_USERNAME`	No	—	Salesforce username
`SALESFORCE_PASSWORD`	No	—	Salesforce password
`SALESFORCE_SECURITY_TOKEN`	No	—	Salesforce security token
`SALESFORCE_CLIENT_ID`	No	—	Salesforce connected app client ID
`SALESFORCE_CLIENT_SECRET`	No	—	Salesforce connected app client secret
`IMAP_HOST`	No	—	IMAP server for email polling
`IMAP_USER`	No	—	IMAP username
`IMAP_PASS`	No	—	IMAP password
`IMAP_FOLDER`	No	`INBOX`	IMAP folder to poll
`FRONTEND_URL`	No	—	Production frontend URL for CORS allowlist

Documentation Index

FILE	WHAT IT COVERS
architecture.md	3-tier hierarchy, capital flow, memory system, message bus, scheduler jobs
agents.md	Every agent: purpose, functions, triggers, DB reads/writes
crm-integrations.md	GHL, HubSpot, Salesforce setup; crm_router API
capital-allocation.md	Kelly criterion, circuit breakers, 72-hour explore protocol
voice-ai.md	Retell + ElevenLabs integration, call flow, post-call processing
data-collection.md	Web scraper, API poller, social signal, pipeline processor
memory-database.md	Full SQLite schema, hot/warm/cold memory, message bus lifecycle
api-reference.md	Every Flask endpoint with method, params, response, curl examples
swarm-board.md	React app: how to run, Board/Overview tabs, board-state.json
setup-guide.md	Step-by-step setup from zero to running
client-onboarding.md	Non-technical guide for solar business owners
cost-tracking.md	API usage table, token tracking, cost projections
troubleshooting.md	Common errors and fixes